Data Brief · Phase 1, DEFINE

Which Yucatán municipalities show groundwater quality flags, and how much of the state does the monitoring network see?

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1. Research question

Between 2012 and 2022, which municipalities of Yucatán have groundwater monitoring sites that CONAGUA flags as unsuitable as a drinking-water source or restricted for irrigation, and how much of the state does the monitoring network actually cover?

2. Objective

Show, with CONAGUA's own quality labels, where the monitored groundwater sites raise a flag, by parameter and by municipality, and measure how much of the state those sites cover. The aim is that a reader can tell where the evidence is solid and where there is none. We describe; we do not explain causes or predict.

3. Audience

Primary: people who decide where to check water or add monitoring (CONAGUA's regional office, state and municipal water utilities). Secondary: residents and community groups who want to know what the public data can and cannot say about their municipality.

4. Relevance

Groundwater covers almost all of the water demand of the three peninsular states (Ríos-Ponce et al., 2020), and in karst terrain dissolved substances reach the aquifer easily (Cupul-Caamal et al.). The challenge notes that measurements are unevenly spread across sites. We made that the second half of the question: a flag only means something where there is a site, so coverage is part of the answer.

5. Unit of analysis

A monitoring site (code `CLAVE`) for one parameter: one record per site and parameter. We then count flags by municipality.

6. Territory

State of Yucatán, at municipality level (CONAGUA region Península de Yucatán). The files hold 147 sites in 70 municipalities: 123 groundwater sites in 65 municipalities and 24 surface-water sites in 11. We analyse groundwater only. Yucatán has 106 municipalities.

7. Period

2012–2022, taken from the sheet name in every file. Each file has one value per site and parameter for the whole period and no sampling dates, so we cannot look at trends or seasons.

8. Sources

CONAGUA, RENAMECA: 30 state-level Excel files, one per parameter, 2,106 rows, with columns N.P., CLAVE, SITIO, ESTADO, MUNICIPIO, RHA, Parámetro, Valor del Parámetro and Calidad. The number of municipalities (106) comes from the INE state profile. References are listed at the end.

9. Variables

Variable	Role in the analysis	Notes
CLAVE, SITIO, MUNICIPIO	Identify the site and where it is	No site has two names or two municipalities.
Calidad (CONAGUA label)	Defines each flag	The label scale is different for each parameter, so labels are never averaged across parameters.
Nitrates, arsenic, fecal coliforms (groundwater file)	Drinking-source flags	Flag = "No apta como FAAP" (nitrates, arsenic) or "Contaminada" (coliforms). FAAP means fuente de abastecimiento de agua potable.
Conductivity, dissolved solids (SDT)	Irrigation flags	Flag = "Dudosa para riego" or "Indeseable para riego" (conductivity); "Cultivos con manejo especial" or "Cultivos tolerantes" (SDT).
Valor del Parámetro	The value behind each label	Stored as text; some read "<0.01" (below detection). Units are not stated in the files.
Left out	Not part of the question	Cadmium, lead and mercury (99–100% below detection, no variation); the file named "Magnesio" (its code is `mn`, manganese, so the name looks wrong); the surface-water files (24 sites, several nearly empty). Alkalinity, hardness, fluoride, iron and chromium stay as context and are not flagged.

10. Initial plan

#	Step	Status / when
1	Merge the 30 files and check row counts	Done (2,106 rows)
2	Profile coverage, empty and censored values, naming problems	Done
3	Separate groundwater from surface sites	Done
4	Flag sites with CONAGUA's labels; count by parameter and by municipality	Evidence Board, Fri 13 Nov, 5 pm
5	Measure coverage: sites per municipality against the 106 municipalities	Evidence Board, Fri 13 Nov, 5 pm
6	Tableau dashboard: municipality view with site counts and flags, a connected coverage view, a filter by use and by parameter, sources and limits on the page	Sat 14 Nov, 12 pm
7	Pitch (5 minutes): question, evidence, and what the data cannot say	Sat 14 Nov, from 1 pm

11. Limits

- No sampling dates, so no trends or seasons. The values are probably summaries of the period, but the files do not say so, and we will not call them averages.
- CONAGUA's labels are used as published. We did not re-test them, and a flag is not a health or crop diagnosis.
- 41 of the 65 monitored municipalities have a single groundwater site, so one site stands for the whole municipality. We report counts, not rates by municipality.
- The network is not a random sample (Mérida alone has 21 of the 123 sites), so results describe the monitored sites, not the whole state.
- The files have no coordinates, so maps can only be drawn by municipality. The analysis is descriptive and makes no causal claims.

AI use

AI use. Claude (Anthropic) was used to write the merge, profiling and analysis scripts, to run the data checks and to help draft this document. The scripts are in the team folder so every number can be reproduced.

References

1. CONAGUA, RENAMECA (Red Nacional de Medición de la Calidad del Agua). 30 state-level Excel files, one per parameter, named `Calidad_de_agua_Estado_<parameter>.xlsx`, sheet "2012-2022".
2. Instituto Nacional de Estadística y Geografía (INEGI) (2021), Panorama sociodemográfico de Yucatán, Censo de Población y Vivienda 2020 (list of the state's municipalities, keys 001 to 106): https://en.www.inegi.org.mx/contenidos/productos/prod_serv/contenidos/espanol/bvinegi/productos/nueva_estruc/702825198046.pdf
3. CONAGUA, Red Nacional de Medición de la Calidad del Agua (RENAMECA), official presentation (groundwater indicators, including manganese, and the quality labels such as "Dudosa para riego" and "Cultivos tolerantes"): https://files.conagua.gob.mx/transparencia/CalidaddelAgua.pdf
4. CONAGUA, Indicadores de Calidad del Agua (groundwater sites are evaluated on suitability as a drinking-water source and for agricultural irrigation), consulted 13 November 2026: https://www.gob.mx/conagua/articulos/indicadores-de-calidad-del-agua
5. Ríos-Ponce, J. D., Acosta-González, G. and Cejudo, E. (2020), La precipitación histórica y la extracción del agua subterránea en la península de Yucatán: una reflexión, Desde el Herbario CICY 12: 110–118: https://cicy.repositorioinstitucional.mx/jspui/bitstream/1003/2588/1/2020-06-04-Rios-Ponce-et-al-La-precipitacion.pdf